

MITRE ATT&CK Campaign Research Report

Intern Details

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Project Details

- Assigned Campaign: C0015
- Software Range: S0150 – S0160

Domain

- Cybersecurity / Threat Intelligence (MITRE ATT&CK)

Executive Summary

This report analyzes the MITRE ATT&CK campaign C0015, focusing on its attack lifecycle, techniques, and tools used by attackers. The study highlights how ransomware campaigns operate in a structured manner and emphasizes detection and mitigation strategies.

The analysis includes software mapping (S0150–S0160), attack flow diagrams, and threat intelligence insights to provide a comprehensive understanding of the campaign.

1. Campaign Research – C0015

C0015 is a ransomware intrusion campaign observed in August 2021.

Attackers used tools such as BazarLoader, Cobalt Strike, and Conti ransomware to gain access, move within the network, and deploy ransomware within a short period.

1.2 Threat Actor / Group

The attack is believed to be associated with the Conti ransomware group based on techniques and tools used during the campaign.

1.3 Timeline

- First Seen: August 2021
- Duration: Short-term campaign (few days)

1.4 Objectives of Attack

- Gain unauthorized access to systems
- Move laterally across the network
- Deploy ransomware to encrypt data
- Demand ransom payment

1.5 Targeted Sectors / Countries

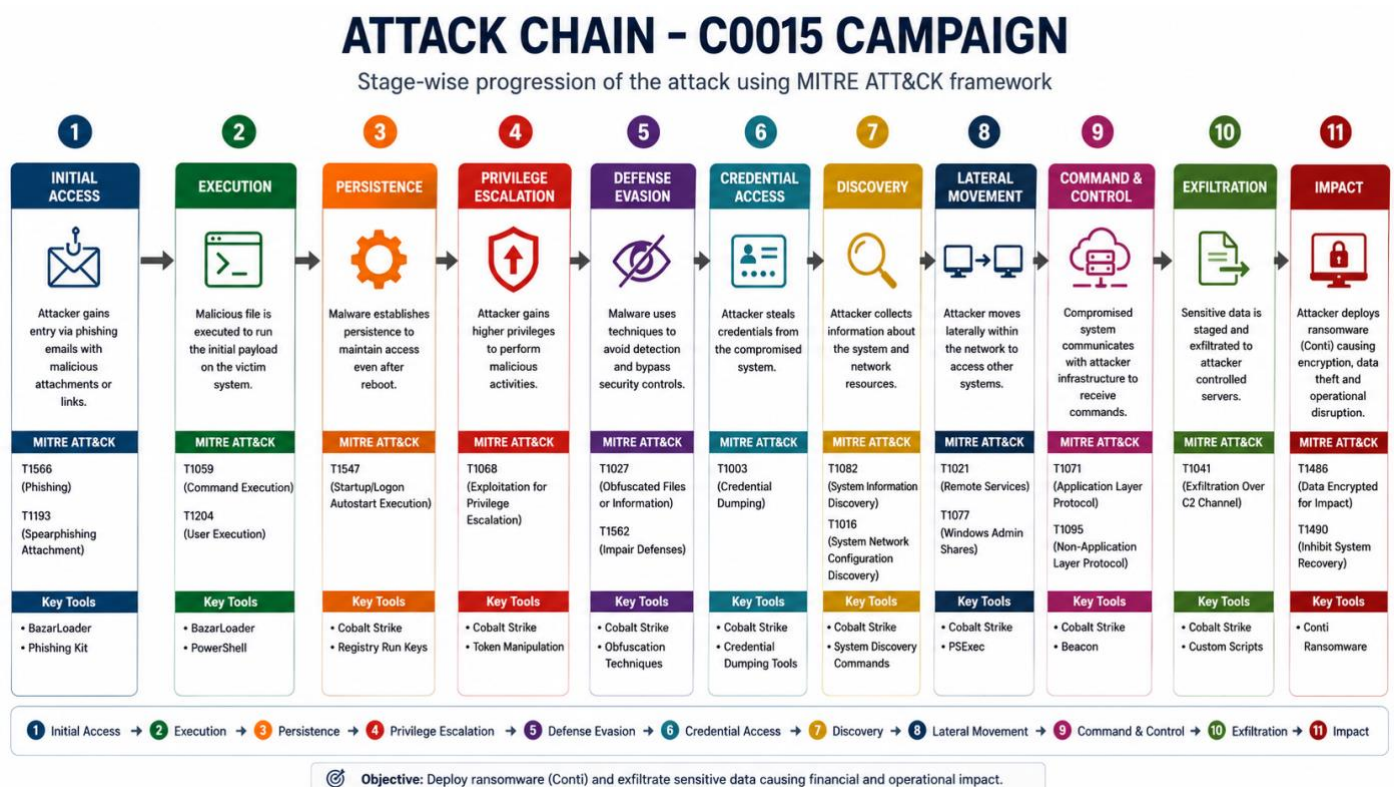
- Likely enterprise organizations
- Targets include business networks
- No specific country confirmed

1.6 Tools Used

- BazarLoader
- Cobalt Strike
- Conti Ransomware

1.7 Attack Flow

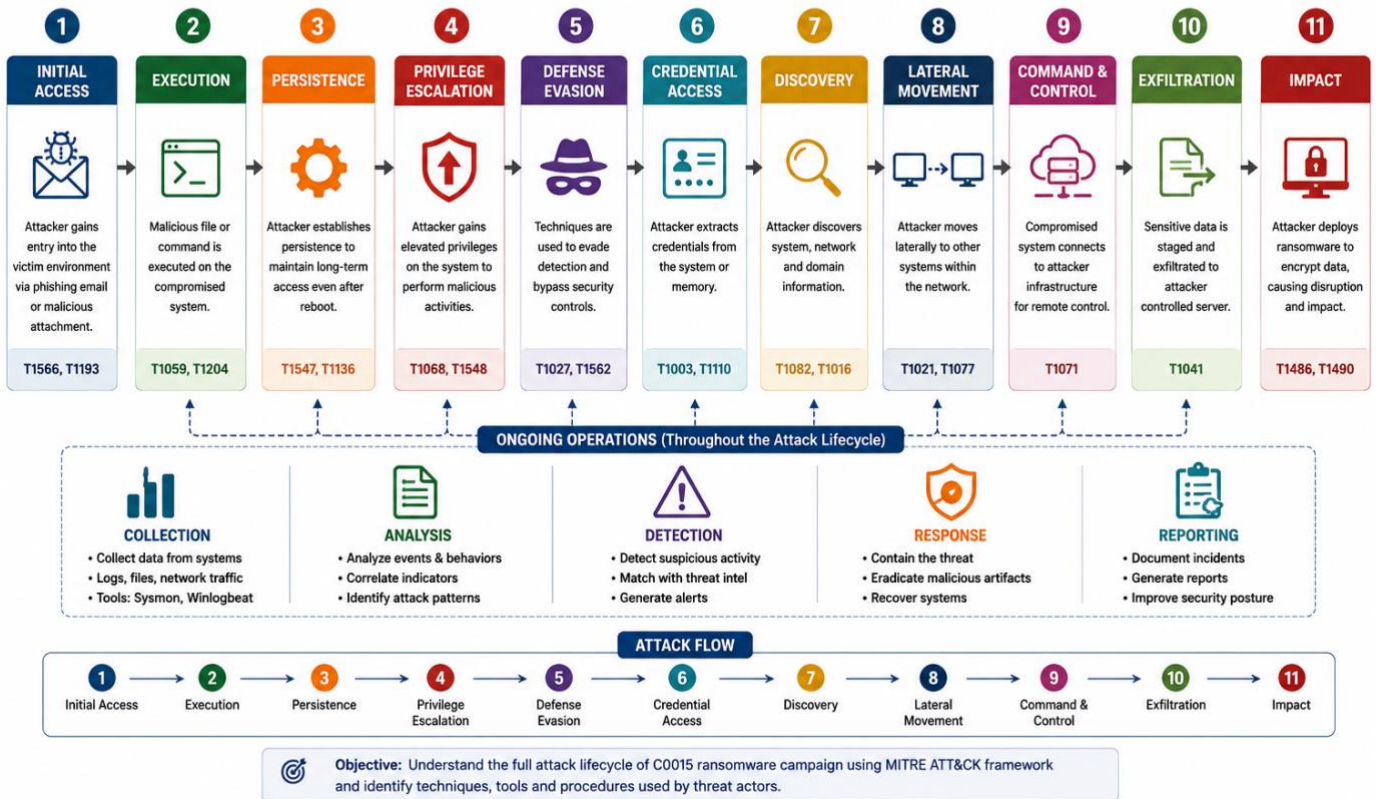
The attack begins with initial access where the attacker enters the system using malware such as BazarLoader. After gaining access, the attacker executes malicious commands to establish control. The attacker then maintains persistence to stay in the system and moves laterally across the network to infect more machines. Finally, ransomware (Conti) is deployed to encrypt data and demand payment.



This is flow-diagram of the Attack Flow, for better understanding.

CYBER ATTACK WORKFLOW (C0015 CAMPAIGN)

Attack Lifecycle based on MITRE ATT&CK Framework



The following workflow illustrates the complete lifecycle of the C0015 attack based on the MITRE ATT&CK framework.

1.8 MITRE ATT&CK Techniques (T-Codes)

- T1059 – Command Execution
Attackers executed malicious commands using scripting tools such as PowerShell.
- T1105 – Ingress Tool Transfer
Additional malicious tools and payloads were transferred into the system after initial compromise.
- T1021 – Remote Services (Lateral Movement)
Attackers moved across systems using remote services to expand control within the network.
- T1078 – Valid Accounts
Compromised or stolen credentials were used to gain legitimate access to systems.

1.9 Real-world Incidents

The C0015 campaign reflects real-world ransomware operations similar to those conducted by the Conti group. Security researchers observed that attackers followed a structured and repeatable attack pattern, often referred to as a “ransomware playbook.”

The rapid execution within a short time frame highlights the efficiency of modern ransomware campaigns.

1.10 Detection Opportunities

- Monitor phishing emails and suspicious attachments (unusual links, macros).
- Detect abnormal PowerShell/command execution on endpoints.
- Look for unexpected outbound connections (unknown IPs/domains).
- Monitor lateral movement patterns (RDP/SMB activity between hosts).
- Identify sudden file encryption activity or mass file changes.
- Check logs for new services / scheduled tasks (persistence signs).

1.11 Defensive Recommendations

- Implement Multi-Factor Authentication (MFA) for all users.
- Apply email filtering & anti-phishing controls.
- Restrict and monitor PowerShell usage (logging, least privilege).
- Use endpoint protection (EDR/antivirus) with real-time monitoring.
- Keep systems patched and updated.
- Enforce least privilege access (limit admin rights).
- Network segmentation to reduce lateral movement.
- Regular backups and test recovery (ransomware readiness).

2. Software Research (S0150 – S0160)

2.1 Software S0150 (POSHSPY)

1. Name & Type

POSHSPY is a **PowerShell-based backdoor malware** used by attackers to maintain hidden access inside compromised systems.

2. Description

POSHSPY has been used by the threat group **APT29** since at least 2015. It acts as a secondary backdoor that helps attackers regain access if their primary malware is removed. The malware uses PowerShell commands to execute malicious activities while remaining difficult to detect.

3. Threat Actor Association

- **APT29 (Cozy Bear)**
 - Associated with advanced cyber espionage campaigns.
-

4. Supported Platforms

- Windows
-

5. MITRE ATT&CK Techniques Used

- **T1059.001 – PowerShell**
Used to execute malicious PowerShell commands.
 - **T1030 – Data Transfer Size Limits**
Uploads stolen data in small chunks to avoid detection.
 - **T1568.002 – Domain Generation Algorithms (DGA)**
Uses dynamically generated domains for communication.
-

6. Execution Method

POSHSPY executes malicious PowerShell scripts within the infected Windows system to run attacker commands.

7. Persistence

The malware maintains long-term access by operating as a hidden secondary backdoor in the compromised environment.

8. Privilege Escalation

POSHSPY may leverage compromised credentials or system weaknesses to gain higher access privileges.

9. Defense Evasion

The malware avoids detection by:

- Using PowerShell instead of traditional executables
 - Splitting transferred data into small chunks
 - Hiding malicious activity within normal system operations
-

10. Credential Access

Attackers may use POSHSPY to maintain access after obtaining credentials from infected systems.

11. Discovery

The malware can gather system information and identify resources available inside the network.

12. Lateral Movement

Attackers may use the established backdoor access to move between systems within the network.

13. Command & Control (C2)

POSHSPY communicates with attacker-controlled infrastructure using generated domains and hidden network communication techniques.

14. Exfiltration

The malware uploads stolen data in small 2048-byte chunks to reduce chances of detection.

15. Impact Analysis

POSHSPY enables persistent unauthorized access, stealthy monitoring, and potential data theft inside compromised enterprise environments.

16. Indicators of Compromise (IOCs)

- Suspicious PowerShell execution
 - Unknown outbound network traffic
 - Unusual PowerShell scripts
 - Hidden communication with external domains
-

17. Detection & Mitigation

- Monitor PowerShell activity
 - Restrict unnecessary PowerShell usage
 - Use endpoint detection and response (EDR) tools
 - Monitor suspicious outbound connections
 - Apply least-privilege access control
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18. References

- MITRE ATT&CK – POSHSPY [MITRE POSHSPY Reference](#)
 - MITRE ATT&CK Framework [MITRE ATT&CK Website](#)
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2.2 Software S0151 (HALFBAKED)

1. Name & Type

HALFBAKED is a backdoor malware used by attackers to maintain remote access and execute commands on compromised systems.

2. Description

HALFBAKED is a lightweight backdoor that enables attackers to remotely control infected systems. It is designed to execute commands, collect information, and maintain persistent access within the target environment. The tool operates stealthily to avoid detection and is commonly used in targeted cyber operations.

3. Threat Actor Association

- Associated with APT1 (Comment Crew)

- Used in cyber espionage and long-term intrusion campaigns
-

4. Supported Platforms

- Windows
-

5. MITRE ATT&CK Techniques Used

- T1059 – Command Execution
Executes commands on the infected system
 - T1105 – Ingress Tool Transfer
Downloads additional tools or payloads
 - T1071 – Application Layer Protocol
Communicates with attacker servers over network protocols
-

6. Execution Method

HALFBAKED executes attacker commands by running processes within the infected system, allowing remote control.

7. Persistence

Maintains access by running continuously in the background and reconnecting to attacker infrastructure.

8. Privilege Escalation

May exploit system weaknesses or credentials to gain elevated privileges.

9. Defense Evasion

- Runs silently in background
 - Minimizes resource usage
 - Avoids detection by security tools
-

10. Credential Access

Supports attackers in accessing system credentials through continued control of the system.

11. Discovery

Collects system information such as processes, files, and network configuration.

12. Lateral Movement

Allows attackers to move within the network by controlling compromised systems.

13. Command & Control (C2)

Communicates with attacker-controlled servers to receive commands and send data.

14. Exfiltration

Can transfer collected data from the system to external servers.

15. Impact Analysis

HALFBAKED enables:

- Persistent unauthorized access
 - Remote system control
 - Data theft
 - Long-term surveillance
-

16. Indicators of Compromise (IOCs)

- Unknown background processes
 - Suspicious outbound connections
 - Unusual system behavior
-

17. Detection & Mitigation

- Monitor system and network activity
 - Use endpoint detection tools
 - Restrict unauthorized processes
 - Apply system updates and patches
-

18. References

- MITRE ATT&CK – HALFBAKED (S0151)
- MITRE ATT&CK Framework

2.3 Software S0152 (EvilGrab)

1. Name & Type

EvilGrab is a **remote access trojan (RAT)** used by attackers to gain full control over compromised systems.

2. Description

EvilGrab is a well-known malware used in cyber espionage campaigns. It allows attackers to remotely control infected machines, capture screenshots, log keystrokes, and steal sensitive data. It communicates with command-and-control (C2) servers to receive instructions and send stolen information.

3. Threat Actor Association

- Associated with **APT groups (especially China-linked groups like APT1 / APT17)**
 - Used in targeted attacks against organizations and government entities
-

4. Supported Platforms

- Windows
-

5. MITRE ATT&CK Techniques Used

- **T1059 – Command Execution**
Executes attacker commands on the system
 - **T1105 – Ingress Tool Transfer**
Downloads additional malicious tools
 - **T1071 – Application Layer Protocol**
Communicates with attacker servers via HTTP/HTTPS
 - **T1041 – Exfiltration Over C2 Channel**
Sends stolen data back to attacker
-

6. Execution Method

EvilGrab executes as a background process after infection, allowing attackers to run commands and control the system remotely.

7. Persistence

Maintains access by installing itself in the system and running automatically on startup.

8. Privilege Escalation

May attempt to gain higher privileges to access sensitive data and system controls.

9. Defense Evasion

- Runs in background silently
 - Uses normal system processes to hide activity
 - Blends with legitimate traffic
-

10. Credential Access

Captures sensitive information such as login credentials through keylogging and system monitoring.

11. Discovery

Collects system details such as files, processes, and network configuration.

12. Lateral Movement

Allows attackers to move across systems by using stolen credentials or remote execution.

13. Command & Control (C2)

Maintains communication with attacker servers to receive instructions and send data.

14. Exfiltration

Transfers stolen data such as credentials, files, and screenshots to attacker-controlled servers.

15. Impact Analysis

EvilGrab enables:

- Full remote control of system
 - Data theft
 - Surveillance (screenshots, keylogging)
 - Long-term unauthorized access
-

16. Indicators of Compromise (IOCs)

- Unusual outbound network traffic
 - Unknown background processes
 - Suspicious file activity
 - Unexpected system behavior
-

17. Detection & Mitigation

- Monitor network traffic for anomalies
 - Use endpoint detection and response (EDR)
 - Restrict unauthorized software execution
 - Keep systems updated
 - Apply strong access controls
-

18. References

- MITRE ATT&CK – EvilGrab (S0152)
- MITRE ATT&CK Framework

2.4 Software S0153 (RedLeaves)

1. Name & Type

RedLeaves is a **remote access trojan (RAT)** used by attackers to gain persistent remote control over compromised systems.

2. Description

RedLeaves is malware used in targeted cyber attacks, especially in espionage campaigns. It allows attackers to execute commands, manage files, collect system information, and maintain long-term access to infected machines. It communicates with command-and-control (C2) servers to receive instructions and send data.

3. Threat Actor Association

- Associated with **APT10 (Stone Panda)**
 - Used in cyber espionage operations targeting organizations and government sectors
-

4. Supported Platforms

- Windows
-

5. MITRE ATT&CK Techniques Used

- **T1059 – Command Execution**
Executes attacker commands on the system
 - **T1105 – Ingress Tool Transfer**
Downloads additional malicious payloads
 - **T1071 – Application Layer Protocol**
Uses HTTP/HTTPS for communication with attacker servers
 - **T1021 – Remote Services (Lateral Movement)**
Enables movement across systems in the network
-

6. Execution Method

RedLeaves runs as a background process after infection, enabling attackers to execute commands remotely on the system.

7. Persistence

Maintains persistence by installing itself in system directories and configuring automatic execution on system startup.

8. Privilege Escalation

May exploit vulnerabilities or use stolen credentials to gain higher-level access.

9. Defense Evasion

- Operates silently in background
 - Uses legitimate processes to hide activity
 - Blends malicious traffic with normal network traffic
-

10. Credential Access

Supports attackers in collecting sensitive information, including user credentials, through system monitoring.

11. Discovery

Collects system and network information such as running processes, files, and configurations.

12. Lateral Movement

Enables attackers to move across systems by leveraging remote services and compromised credentials.

13. Command & Control (C2)

Communicates with attacker-controlled servers to receive commands and send collected data.

14. Exfiltration

Transfers stolen data such as files and credentials to external servers.

15. Impact Analysis

RedLeaves allows attackers to:

- Maintain persistent access
 - Perform remote system control
 - Steal sensitive data
 - Conduct espionage activities
-

16. Indicators of Compromise (IOCs)

- Suspicious outbound network traffic
 - Unknown background processes
 - Unexpected system behavior
 - Unauthorized file modifications
-

17. Detection & Mitigation

- Monitor abnormal network communication
 - Use endpoint detection tools (EDR)
 - Restrict unauthorized program execution
 - Apply security patches regularly
 - Enforce strong access control policies
-

18. References

- MITRE ATT&CK – RedLeaves (S0153)
- MITRE ATT&CK Framework

2.5 Software S0154 (Cobalt Strike)

1. Name & Type

Cobalt Strike is a post-exploitation framework / penetration testing tool that is widely abused by attackers for command execution, lateral movement, and persistence.

2. Description

Cobalt Strike is originally a legitimate security testing tool used by penetration testers. However, it is frequently used by attackers as a powerful post-exploitation toolkit. It provides capabilities such as remote command execution, credential theft, lateral movement, and communication with command-and-control (C2) servers through its “Beacon” payload.

3. Threat Actor Association

- Used by multiple APT groups and ransomware operators
 - Commonly seen in campaigns involving Conti, APT29, APT10, and other advanced threat actors
-

4. Supported Platforms

- Windows
 - (Also supports cross-platform operations through payloads)
-

5. MITRE ATT&CK Techniques Used

- T1059 – Command Execution
Executes commands on compromised systems
- T1105 – Ingress Tool Transfer
Transfers additional tools and payloads
- T1021 – Remote Services (Lateral Movement)
Enables movement across network systems
- T1071 – Application Layer Protocol
Communicates with C2 servers over HTTP/HTTPS
- T1003 – Credential Dumping
Extracts credentials from system memory

6. Execution Method

Cobalt Strike deploys a payload called Beacon, which runs on the target system and allows attackers to execute commands and control the system remotely.

7. Persistence

Maintains access using techniques like scheduled tasks, startup entries, or registry modifications.

8. Privilege Escalation

Can escalate privileges by exploiting system vulnerabilities or using stolen credentials.

9. Defense Evasion

- Uses encrypted communication
 - Mimics normal network traffic
 - Can operate in memory to avoid detection
-

10. Credential Access

Supports credential dumping and harvesting from compromised systems.

11. Discovery

Performs system and network discovery to identify targets for further exploitation.

12. Lateral Movement

Enables attackers to move within networks using remote execution and stolen credentials.

13. Command & Control (C2)

Communicates with attacker servers using flexible and encrypted channels, often disguised as normal traffic.

14. Exfiltration

Allows attackers to transfer sensitive data from compromised systems.

15. Impact Analysis

Cobalt Strike enables:

- Full system control
 - Network-wide compromise
 - Data theft
 - Advanced attack operations
-

16. Indicators of Compromise (IOCs)

- Suspicious network traffic patterns
 - Beacon-like communication intervals
 - Unknown processes in memory
 - Abnormal command execution
-

17. Detection & Mitigation

- Monitor network traffic for anomalies
 - Detect Beacon behavior patterns
 - Use EDR solutions
 - Restrict unauthorized remote execution
 - Apply strong access controls
-

18. References

- MITRE ATT&CK – Cobalt Strike (S0154)
- MITRE ATT&CK Framework

2.6 Software S0155 (WINDSHIELD)

1. Name & Type

WINDSHIELD is a **backdoor malware** used by attackers to maintain persistent access and remotely control compromised systems.

2. Description

WINDSHIELD is a backdoor tool used in targeted cyber attacks to provide attackers with long-term access to infected systems. It allows execution of commands, system monitoring, and communication with command-and-control (C2) servers. The malware is designed to operate stealthily and support multiple stages of an attack, including persistence and data collection.

3. Threat Actor Association

- Associated with **APT groups (notably APT1 / similar espionage groups)**
 - Used in targeted cyber espionage operations
-

4. Supported Platforms

- Windows
-

5. MITRE ATT&CK Techniques Used

- **T1059 – Command Execution**
Executes attacker commands on the system
 - **T1105 – Ingress Tool Transfer**
Downloads additional tools or payloads
 - **T1071 – Application Layer Protocol**
Communicates with attacker servers via HTTP/HTTPS
 - **T1041 – Exfiltration Over C2 Channel**
Transfers stolen data to attacker infrastructure
-

6. Execution Method

WINDSHIELD executes as a background process on the infected system, enabling attackers to run commands remotely.

7. Persistence

Maintains persistence by ensuring continuous execution within the system, often through startup mechanisms or hidden processes.

8. Privilege Escalation

May leverage system vulnerabilities or credentials to gain higher-level access.

9. Defense Evasion

- Runs silently in background
 - Blends with legitimate system processes
 - Minimizes detection by security tools
-

10. Credential Access

Supports attackers by maintaining access and enabling further credential harvesting operations.

11. Discovery

Collects system and network information such as files, processes, and configurations.

12. Lateral Movement

Allows attackers to move across systems by using remote execution and established access.

13. Command & Control (C2)

Communicates with attacker-controlled servers to receive commands and send collected data.

14. Exfiltration

Transfers sensitive data from compromised systems to external servers.

15. Impact Analysis

WINDSHIELD enables:

- Persistent unauthorized access
 - Remote system control
 - Data theft
 - Long-term espionage
-

16. Indicators of Compromise (IOCs)

- Suspicious background processes
 - Unusual outbound network traffic
 - Unknown connections to external servers
 - Unexpected system behavior
-

17. Detection & Mitigation

- Monitor system and network activity
 - Use endpoint detection tools (EDR)
 - Restrict unauthorized processes
 - Apply security updates and patches
 - Enforce least privilege access
-

18. References

- MITRE ATT&CK – WINDSHIELD (S0155)
- MITRE ATT&CK Framework

2.7 Software S0156 (KOMPROGO)

1. Name & Type

KOMPROGO is a remote access trojan (RAT) / backdoor malware used by attackers to gain and maintain remote control over compromised systems.

2. Description

KOMPROGO is a backdoor tool used in targeted cyber operations, particularly in espionage campaigns. It enables attackers to execute commands, collect system information, and communicate with command-and-control (C2) servers. The malware is designed to maintain persistence and support long-term unauthorized access.

3. Threat Actor Association

- Associated with APT10 (Stone Panda)
 - Used in cyber espionage campaigns targeting organizations and enterprises
-

4. Supported Platforms

- Windows
-

5. MITRE ATT&CK Techniques Used

- T1059 – Command Execution
Executes attacker commands on the system
 - T1105 – Ingress Tool Transfer
Downloads additional payloads or tools
 - T1071 – Application Layer Protocol
Communicates with C2 servers using HTTP/HTTPS
 - T1021 – Remote Services (Lateral Movement)
Allows movement across systems in the network
-

6. Execution Method

KOMPROGO runs as a background process after infection and allows attackers to execute commands remotely.

7. Persistence

Maintains persistence by embedding itself into the system and ensuring continued execution.

8. Privilege Escalation

May exploit vulnerabilities or use stolen credentials to gain elevated privileges.

9. Defense Evasion

- Operates silently in background
 - Uses legitimate system processes to hide activity
 - Blends malicious traffic with normal network communication
-

10. Credential Access

Supports attackers in obtaining or using credentials for further access.

11. Discovery

Collects system and network information such as processes, files, and configuration details.

12. Lateral Movement

Enables attackers to move across systems using remote execution and credentials.

13. Command & Control (C2)

Communicates with attacker-controlled servers to receive commands and send data.

14. Exfiltration

Transfers stolen data from compromised systems to external servers.

15. Impact Analysis

KOMPROGO enables:

- Persistent unauthorized access
 - Remote system control
 - Data theft
 - Long-term espionage activities
-

16. Indicators of Compromise (IOCs)

- Suspicious outbound network traffic
 - Unknown background processes
 - Unusual system behavior
 - Unauthorized connections
-

17. Detection & Mitigation

- Monitor abnormal network activity
 - Use endpoint detection and response (EDR)
 - Restrict unauthorized execution
 - Apply security patches
 - Enforce access control policies
-

18. References

- MITRE ATT&CK – KOMPROGO (S0156)
- MITRE ATT&CK Framework

2.8 Software S0157 (SOUNDBITE)

1. Name & Type

SOUNDBITE is a backdoor malware used by attackers to maintain remote access and execute commands on compromised systems.

2. Description

SOUNDBITE is a malware tool used in targeted cyber operations, particularly in espionage campaigns. It allows attackers to execute commands, gather system information, and maintain persistent access to infected systems. The malware communicates with command-and-control (C2) infrastructure to receive instructions and send data.

3. Threat Actor Association

- Associated with APT10 (Stone Panda)
 - Used in long-term cyber espionage campaigns targeting organizations
-

4. Supported Platforms

- Windows

5. MITRE ATT&CK Techniques Used

- T1059 – Command Execution
Executes commands on the infected system
- T1105 – Ingress Tool Transfer
Downloads additional payloads or tools
- T1071 – Application Layer Protocol
Communicates with attacker servers over HTTP/HTTPS
- T1021 – Remote Services (Lateral Movement)
Enables movement across systems

6. Execution Method

SOUNDBITE runs as a background process after infection and allows attackers to execute commands remotely.

7. Persistence

Maintains persistence by ensuring continued execution through system processes or startup mechanisms.

8. Privilege Escalation

May use vulnerabilities or compromised credentials to gain elevated privileges.

9. Defense Evasion

- Operates silently in background
- Uses legitimate system processes
- Blends with normal system activity

10. Credential Access

Supports attackers in accessing or using system credentials for further compromise.

11. Discovery

Collects information about system configuration, processes, and network environment.

12. Lateral Movement

Enables attackers to move within the network using remote execution methods.

13. Command & Control (C2)

Communicates with attacker-controlled servers to receive commands and send collected data.

14. Exfiltration

Transfers sensitive data from compromised systems to external servers.

15. Impact Analysis

SOUNDBITE enables:

- Persistent system compromise
 - Remote control of systems
 - Data theft
 - Long-term espionage
-

16. Indicators of Compromise (IOCs)

- Suspicious outbound network traffic
 - Unknown background processes
 - Unusual system behavior
 - Unauthorized external connections
-

17. Detection & Mitigation

- Monitor network traffic for anomalies
 - Use endpoint detection tools (EDR)
 - Restrict unauthorized execution
 - Apply security updates
 - Enforce strong access control
-

18. References

- MITRE ATT&CK – SOUNDBITE (S0157)
- MITRE ATT&CK Framework

2.9 Software S0158 (PHOREAL)

1. Name & Type

PHOREAL is a backdoor malware / remote access tool used by attackers to maintain persistent access and control compromised systems.

2. Description

PHOREAL is a malware tool used in targeted cyber operations, especially espionage campaigns. It allows attackers to execute commands, gather system information, and communicate with command-and-control (C2) servers. The malware is designed to support long-term access and stealthy operations within compromised environments.

3. Threat Actor Association

- Associated with APT10 (Stone Panda)
 - Used in advanced cyber espionage campaigns targeting organizations
-

4. Supported Platforms

- Windows
-

5. MITRE ATT&CK Techniques Used

- T1059 – Command Execution
Executes commands on the system
 - T1105 – Ingress Tool Transfer
Downloads additional payloads or tools
 - T1071 – Application Layer Protocol
Communicates with attacker servers via HTTP/HTTPS
 - T1021 – Remote Services (Lateral Movement)
Enables movement across systems
-

6. Execution Method

PHOREAL executes as a background process after infection, allowing attackers to run commands remotely.

7. Persistence

Maintains persistence by ensuring continuous execution within the system, often through hidden processes or startup mechanisms.

8. Privilege Escalation

May exploit vulnerabilities or use stolen credentials to gain higher-level access.

9. Defense Evasion

- Runs silently in background
 - Uses legitimate processes to hide activity
 - Blends malicious communication with normal traffic
-

10. Credential Access

Supports attackers in accessing or using system credentials for further compromise.

11. Discovery

Collects system and network information such as processes, files, and configurations.

12. Lateral Movement

Allows attackers to move within the network using remote execution and credentials.

13. Command & Control (C2)

Communicates with attacker-controlled servers to receive commands and send collected data.

14. Exfiltration

Transfers stolen data from compromised systems to external servers.

15. Impact Analysis

PHOREAL enables:

- Persistent unauthorized access
 - Remote system control
 - Data theft
 - Long-term espionage activities
-

16. Indicators of Compromise (IOCs)

- Suspicious outbound network traffic
 - Unknown background processes
 - Unusual system behavior
 - Unauthorized external connections
-

17. Detection & Mitigation

- Monitor network traffic for anomalies
 - Use endpoint detection and response (EDR)
 - Restrict unauthorized execution
 - Apply system patches
 - Enforce least privilege access
-

18. References

- MITRE ATT&CK – PHOREAL (S0158)
- MITRE ATT&CK Framework

2.10 Software S0159 (SNUGRIDE)

1. Name & Type

SNUGRIDE is a **backdoor malware / remote access tool** used to maintain persistent access and execute commands on compromised systems.

2. Description

SNUGRIDE is a malware tool used in targeted cyber espionage campaigns. It enables attackers to remotely control infected systems, execute commands, collect system information, and communicate with command-and-control (C2) servers. It is designed for stealthy and long-term operations within compromised environments.

3. Threat Actor Association

- Associated with **APT10 (Stone Panda)**
 - Used in cyber espionage campaigns targeting organizations and enterprise networks
-

4. Supported Platforms

- Windows
-

5. MITRE ATT&CK Techniques Used

- **T1059 – Command Execution**
Executes attacker commands on the system
 - **T1105 – Ingress Tool Transfer**
Downloads additional payloads
 - **T1071 – Application Layer Protocol**
Communicates with attacker servers via HTTP/HTTPS
 - **T1021 – Remote Services (Lateral Movement)**
Enables movement across systems
-

6. Execution Method

SNUGRIDE runs as a background process and allows attackers to execute commands remotely.

7. Persistence

Maintains persistence by ensuring continuous execution through system processes or startup mechanisms.

8. Privilege Escalation

May exploit vulnerabilities or use stolen credentials to gain elevated access.

9. Defense Evasion

- Operates silently in background
 - Uses legitimate system processes
 - Blends with normal network traffic
-

10. Credential Access

Supports attackers in accessing or using credentials for further compromise.

11. Discovery

Collects system and network information such as files, processes, and configurations.

12. Lateral Movement

Enables attackers to move across systems using remote execution and credentials.

13. Command & Control (C2)

Communicates with attacker-controlled servers to receive commands and send data.

14. Exfiltration

Transfers stolen data from compromised systems to external servers.

15. Impact Analysis

SNUGRIDE enables:

- Persistent system compromise
 - Remote control of systems
 - Data theft
 - Long-term espionage
-

16. Indicators of Compromise (IOCs)

- Suspicious outbound connections
 - Unknown background processes
 - Unusual system behavior
 - Unauthorized communication with external servers
-

17. Detection & Mitigation

- Monitor abnormal network activity
 - Use endpoint detection tools (EDR)
 - Restrict unauthorized execution
 - Apply security updates
 - Enforce strong access control policies
-

18. References

- MITRE ATT&CK – SNUGRIDE (S0159)
- MITRE ATT&CK Framework

2.11 Software S0160 (CERTUTIL)

1. Name & Type

certutil is a **legitimate Windows command-line utility** that is often **abused by attackers** (**Living-off-the-Land tool**).

2. Description

certutil is a built-in Windows tool used for managing certificates. Attackers misuse it to download malicious files, encode/decode data, and transfer payloads without raising suspicion. Because it is a trusted system tool, it helps attackers bypass security detection.

3. Threat Actor Association

- Used by **multiple APT groups and ransomware operators**
 - Common in real-world attacks due to its built-in nature
-

4. Supported Platforms

- Windows

5. MITRE ATT&CK Techniques Used

- **T1105 – Ingress Tool Transfer**
Used to download malicious files from remote servers
- **T1059 – Command Execution**
Executes commands through command-line interface
- **T1140 – Deobfuscate/Decode Files or Information**
Decodes encoded payloads
- **T1027 – Obfuscated Files or Information**
Used to hide malicious content

6. Execution Method

certutil is executed through command-line to download or manipulate files from remote sources.

7. Persistence

Not directly used for persistence, but supports other tools in maintaining access.

8. Privilege Escalation

Not directly used, but may assist in post-exploitation activities.

9. Defense Evasion

- Uses legitimate Windows tool
- Avoids detection by blending with normal activity
- Bypasses application whitelisting

10. Credential Access

Not directly used for credential access.

11. Discovery

Not primarily used for discovery, but supports other stages.

12. Lateral Movement

Not directly used, but may assist in payload transfer between systems.

13. Command & Control (C2)

Used to download payloads from attacker-controlled servers.

14. Exfiltration

Can encode and transfer data, aiding in data exfiltration.

15. Impact Analysis

certutil enables:

- Stealthy payload delivery
 - Bypassing security controls
 - Supporting larger attack chains
-

16. Indicators of Compromise (IOCs)

- Unusual certutil command usage
 - Downloads from unknown URLs
 - Suspicious command-line activity
-

17. Detection & Mitigation

- Monitor command-line usage
 - Restrict unnecessary certutil execution
 - Use endpoint monitoring tools
 - Analyze unusual file downloads
-

18. References

- MITRE ATT&CK – certutil (S0160)
- MITRE ATT&CK Framework

Tool-to-Attack Stage Mapping (C0015 Campaign)

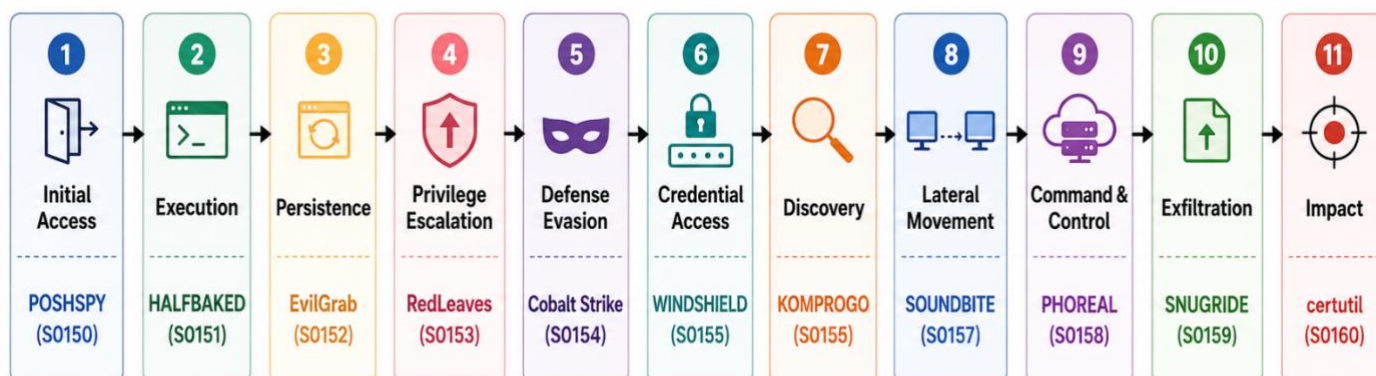


Figure: Mapping of Tools (S0150–S0160) to Attack Stages in C0015 Campaign

The following diagram maps each analyzed tool (S0150–S0160) to its corresponding stage in the attack lifecycle.

Key Takeaways

- Modern attacks follow structured multi-stage lifecycles
- Attackers use multiple tools for persistence and stealth
- Early detection is critical to prevent ransomware impact

Attack Progression:

Initial Access → Execution → Persistence → Lateral Movement → Exfiltration → Impact

➤ Final Summary

The C0015 campaign is a ransomware attack where attackers follow a structured step-by-step approach to compromise systems. They begin by gaining initial access using malware, then execute commands to establish control and maintain persistence within the network.

Attackers then move laterally across systems, collect credentials and sensitive information, and prepare the environment for maximum impact. Finally, ransomware is deployed to encrypt data and demand payment.

This campaign highlights that modern cyber attacks are multi-stage, well-planned, and focused on maximizing damage. Early detection at initial access or lateral movement stages is critical to prevent major impact.

